

This is CS50

reading levels

One fish. Two fish. Red fish. Blue fish.

Before Grade 1

Congratulations! Today is your day. You're off to Great
Places! You're off and away!

Grade 3

It was a bright cold day in April, and the clocks were striking thirteen. Winston Smith, his chin nuzzled into his breast in an effort to escape the vile wind, slipped quickly through the glass doors of Victory Mansions, though not quickly enough to prevent a swirl of gritty dust from entering along with him.

Grade 10

reading levels

cryptography

U I J T J T D T 5 0

T H I S I S C S 5 0



debugging





9/2

9/9

0800 Antan started
 1000 " stopped - antan ✓ { 1.2700 9.037 847 025
 1300 (032) MP-MC ~~1.582142000~~ 9.037 846 895 correct
 (033) PRO 2 2.130476415 (2) 4.615925059(-2)
 correct 2.130676415
 Relays 6-2 in 033 failed special speed test
 in relay " 10.000 test.

Relays changed
 1100 Started Cosine Tape (Sine check)
 1525 Started Multi-Adder Test.

1545



Relay #70 Panel F
 (moth) in relay.

First actual case of bug being found.
 1630 Antan started.
 1700 closed down.

Relay
 3145
 Relay 337

In relay

11,000 test.

1100 Started Cosine Tapc (Sine check)
1525 Started Multi-Adder Test.

1545



Relay #70 Panel F
(moth) in relay.

First actual case of bug being found.
~~1630~~ 1630 Antangut started.
1700 closed down.

printf

printf

debug50

printf

debug50

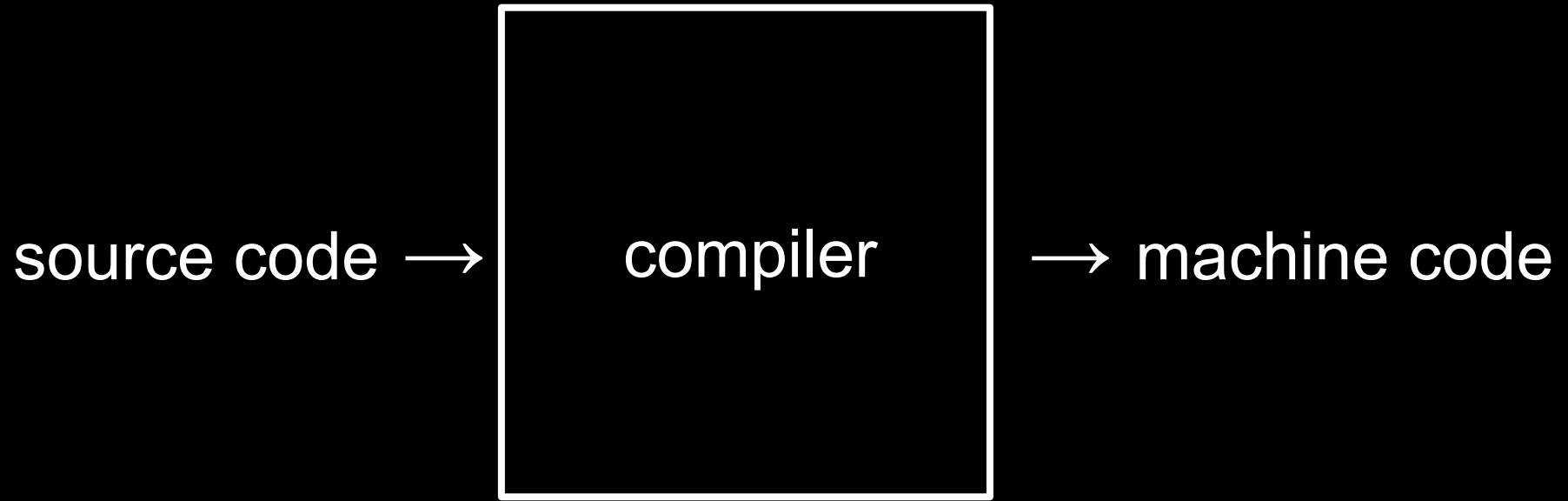
rubber duck

printf

debug50

rubber duck

CS50 Duck



```
#include <stdio.h>
```

```
int main(void)
```

```
{
```

```
    printf("hello, world\n");
```

```
}
```

01111111	01000101	01001100	01000110	00000010	00000001	00000001	00000000
00000000	00000000	00000000	00000000	00000000	00000000	00000000	00000000
00000010	00000000	00111110	00000000	00000001	00000000	00000000	00000000
10110000	00000101	01000000	00000000	00000000	00000000	00000000	00000000
01000000	00000000	00000000	00000000	00000000	00000000	00000000	00000000
11010000	00010011	00000000	00000000	00000000	00000000	00000000	00000000
00000000	00000000	00000000	00000000	01000000	00000000	00111000	00000000
00001001	00000000	01000000	00000000	00100100	00000000	00100001	00000000
00000110	00000000	00000000	00000000	00000101	00000000	00000000	00000000
01000000	00000000	00000000	00000000	00000000	00000000	00000000	00000000
01000000	00000000	01000000	00000000	00000000	00000000	00000000	00000000
01000000	00000000	01000000	00000000	00000000	00000000	00000000	00000000
11111000	00000001	00000000	00000000	00000000	00000000	00000000	00000000
11111000	00000001	00000000	00000000	00000000	00000000	00000000	00000000
00001000	00000000	00000000	00000000	00000000	00000000	00000000	00000000
00000011	00000000	00000000	00000000	00000100	00000000	00000000	00000000
00111000	00000010	00000000	00000000	00000000	00000000	00000000	00000000

...

```
make hello
```

```
./hello
```

```
clang hello.c
```

```
./a.out
```

command-line arguments

```
clang -o hello hello.c
```

```
./hello
```

```
#include <stdio.h>
```

```
int main(void)  
{  
    printf("hello, world\n");  
}
```

```
#include <cs50.h>
#include <stdio.h>

int main(void)
{
    string name = get_string("What's your name? ");
    printf("hello, %s\n", name);
}
```

```
clang -o hello hello.c -lcs50
```

```
./hello
```

```
make hello
```

```
./hello
```

compiling

preprocessing

compiling

assembling

linking

preprocessing

compiling

assembling

linking

```
#include <stdio.h>
```

```
void meow(void);
```

```
int main(void)
```

```
{  
    for (int i = 0; i < 3; i++)  
    {  
        meow();  
    }  
}
```

```
void meow(void)
```

```
{  
    printf("meow\n");  
}
```

```
#include <stdio.h>
```

```
void meow(void);
```

```
int main(void)
{
    for (int i = 0; i < 3; i++)
    {
        meow();
    }
}
```

```
void meow(void)
{
    printf("meow\n");
}
```

```
#include <cs50.h>
#include <stdio.h>

int main(void)
{
    string name = get_string("What's your name? ");
    printf("hello, %s\n", name);
}
```

```
#include <cs50.h>
#include <stdio.h>
```

```
int main(void)
{
    string name = get_string("What's your name? ");
    printf("hello, %s\n", name);
}
```

```
string get_string(string prompt);
```

```
#include <stdio.h>
```

```
int main(void)
```

```
{
```

```
    string name = get_string("What's your name? ");
```

```
    printf("hello, %s\n", name);
```

```
}
```

```
string get_string(string prompt);
```

```
#include <stdio.h>
```

```
int main(void)
```

```
{
```

```
    string name = get_string("What's your name? ");
```

```
    printf("hello, %s\n", name);
```

```
}
```

```
string get_string(string prompt);  
int printf(string format, ...);
```

```
int main(void)  
{  
    string name = get_string("What's your name? ");  
    printf("hello, %s\n", name);  
}
```

preprocessing

compiling

assembling

linking

```
string get_string(string prompt);  
int printf(string format, ...);  
  
int main(void)  
{  
    string name = get_string("What's your name? ");  
    printf("hello, %s\n", name);  
}
```

```

...
main:                                # @main
    .cfi_startproc
# BB#0:
    pushq    %rbp
.Ltmp0:
    .cfi_def_cfa_offset 16
.Ltmp1:
    .cfi_offset %rbp, -16
    movq     %rsp, %rbp
.Ltmp2:
    .cfi_def_cfa_register %rbp
    subq     $16, %rsp
    xorl     %eax, %eax
    movl     %eax, %edi
    movabsq   $.L.str, %rsi
    movb     $0, %al
    callq    get_string
    movabsq   $.L.str.1, %rdi
    movq     %rax, -8(%rbp)
    movq     -8(%rbp), %rsi
    movb     $0, %al
    callq    printf
...

```

```

...
main:                                     # @main
    .cfi_startproc
# BB#0:
    pushq    %rbp
.Ltmp0:
    .cfi_def_cfa_offset 16
.Ltmp1:
    .cfi_offset %rbp, -16
    movq     %rsp, %rbp
.Ltmp2:
    .cfi_def_cfa_register %rbp
    subq     $16, %rsp
    xorl     %eax, %eax
    movl     %eax, %edi
    movabsq   $.L.str, %rsi
    movb     $0, %al
    callq    get_string
    movabsq   $.L.str.1, %rdi
    movq     %rax, -8(%rbp)
    movq     -8(%rbp), %rsi
    movb     $0, %al
    callq    printf
...

```

```
...
main:                                # @main
    .cfi_startproc
# BB#0:
    pushq    %rbp
.Ltmp0:
    .cfi_def_cfa_offset 16
.Ltmp1:
    .cfi_offset %rbp, -16
    movq     %rsp, %rbp
.Ltmp2:
    .cfi_def_cfa_register %rbp
    subq     $16, %rsp
    xorl     %eax, %eax
    movl     %eax, %edi
    movabsq  $.L.str, %rsi
    movb     $0, %al
    callq    get_string
    movabsq  $.L.str.1, %rdi
    movq     %rax, -8(%rbp)
    movq     -8(%rbp), %rsi
    movb     $0, %al
    callq    printf
...
```

preprocessing

compiling

assembling

linking

```

...
main:                                # @main
    .cfi_startproc
# BB#0:
    pushq    %rbp
.Ltmp0:
    .cfi_def_cfa_offset 16
.Ltmp1:
    .cfi_offset %rbp, -16
    movq     %rsp, %rbp
.Ltmp2:
    .cfi_def_cfa_register %rbp
    subq     $16, %rsp
    xorl     %eax, %eax
    movl     %eax, %edi
    movabsq   $.L.str, %rsi
    movb     $0, %al
    callq    get_string
    movabsq   $.L.str.1, %rdi
    movq     %rax, -8(%rbp)
    movq     -8(%rbp), %rsi
    movb     $0, %al
    callq    printf
...

```

01111111010001010100110001000110
00000010000000010000000100000000
00000000000000000000000000000000
00000000000000000000000000000000
00000001000000000011111000000000
00000001000000000000000000000000
00000000000000000000000000000000
00000000000000000000000000000000
00000000000000000000000000000000
00000000000000000000000000000000
00000000000000000000000000000000
10100000000000100000000000000000
00000000000000000000000000000000
00000000000000000000000000000000
01000000000000000000000000000000
00000000000000001000000000000000
00001010000000000000001000000000
01010101010010001000100111100101
0100100010000011111011000010000
00110001110000001000100111000111
01001000101111100000000000000000
00000000000000000000000000000000
00000000000000001011000000000000
11101000000000000000000000000000
00000000010010001011111000000000
00000000000000000000000000000000
0000000000000000000000001001000

...

preprocessing

compiling

assembling

linking

```
#include <cs50.h>
#include <stdio.h>

int main(void)
{
    string name = get_string("What's your name? ");
    printf("hello, %s\n", name);
}
```

hello.c

hello.c

cs50.c

hello.c

cs50.c

stdio.c

01111111010001010100110001000110
00000010000000010000000100000000
00000000000000000000000000000000
00000000000000000000000000000000
00000001000000000011111000000000
00000001000000000000000000000000
00000000000000000000000000000000
00000000000000000000000000000000
00000000000000000000000000000000
00000000000000000000000000000000
00000000000000000000000000000000
10100000000000100000000000000000
00000000000000000000000000000000
00000000000000000000000000000000
01000000000000000000000000000000
00000000000000001000000000000000
00001010000000000000001000000000
01010101010010001000100111100101
01001000100000111110110000010000
00110001110000001000100111000111
01001000101111100000000000000000
00000000000000000000000000000000
00000000000000000101100000000000
11101000000000000000000000000000
00000000010010001011111000000000
00000000000000000000000000000000
0000000000000000000000001001000

cs50.c

stdio.c

...

01111111010001010100110001000110
00000010000000010000000100000000
00000000000000000000000000000000
00000000000000000000000000000000
00000001000000000011111000000000
00000001000000000000000000000000
00000000000000000000000000000000
00000000000000000000000000000000
00000000000000000000000000000000
00000000000000000000000000000000
10100000000000100000000000000000
00000000000000000000000000000000
00000000000000000000000000000000
01000000000000000000000000000000
00000000000000001000000000000000
0000101000000000000000100000000
01010101010010001000100111100101
01001000100000111110110000010000
0011000111000001000100111000111
0100100010111110000000000000000
0000000000000000000000000000000
0000000000000000101100000000000
1110100000000000000000000000000
0000000001001000101111100000000
0000000000000000000000000000000
0000000000000000000000001001000

01111111010001010100110001000110
00000010000000010000000100000000
00000000000000000000000000000000
00000000000000000000000000000000
00000011000000000011111000000000
00000001000000000000000000000000
11000000000011110000000000000000
00000000000000000000000000000000
01000000000000000000000000000000
00000000000000000000000000000000
00101000001100100000000000000000
00000000000000000000000000000000
00000000000000000000000000000000
01000000000000000011100000000000
00000111000000000100000000000000
00011100000000000000110010000000
00000001000000000000000000000000
00000101000000000000000000000000
00000000000000000000000000000000
00000000000000000000000000000000
00000000000000000000000000000000
00000000000000000000000000000000
01011100001001010000000000000000
00000000000000000000000000000000

stdio.c

...

...

01111111010001010100110001000110
00000010000000010000000100000000
00000000000000000000000000000000
00000000000000000000000000000000
00000001000000000011111000000000
00000001000000000000000000000000
00000000000000000000000000000000
00000000000000000000000000000000
00000000000000000000000000000000
00000000000000000000000000000000
10100000000000100000000000000000
00000000000000000000000000000000
00000000000000000000000000000000
01000000000000000000000000000000
00000000000000001000000000000000
00001010000000000000001000000000
01010101010010001000100111100101
01001000100000111110110000010000
0011000111000001000100111000111
01001000101111100000000000000000
00000000000000000000000000000000
00000000000000001011000000000000
11101000000000000000000000000000
0000000001001000101111100000000
00000000000000000000000000000000
0000000000000000000000001001000

...

01111111010001010100110001000110
00000010000000010000000100000000
00000000000000000000000000000000
00000000000000000000000000000000
00000011000000000011111000000000
00000001000000000000000000000000
11000000000011110000000000000000
00000000000000000000000000000000
01000000000000000000000000000000
00000000000000000000000000000000
00101000001100100000000000000000
00000000000000000000000000000000
00000000000000000000000000000000
01000000000000000111000000000000
00000111000000000100000000000000
00011100000000000001100100000000
00000001000000000000000000000000
00000101000000000000000000000000
00000000000000000000000000000000
00000000000000000000000000000000
00000000000000000000000000000000
00000000000000000000000000000000
01011100001001010000000000000000
00000000000000000000000000000000

...

00101111011011000110100101100010
01100011001011100111001101101111
00101110001101100010000000101111
01110101011100110111001000101111
01101100011010010110001000101111
01111000001110000011011001011111
00110110001101000010110101101100
01101001011011100111010101111000
00101101011001110110111001110101
00101111011011000110100101100010
01100011010111110110111001101111
01101110011100110110100001100001
01110010011001010110010000101110
01100001001000000010000001000001
01010011010111110100111001000101
01000101010001000100010101000100
00100000001010000010000000101111
01101100011010010110001000101111
01111000001110000011011001011111
00110110001101000010110101101100
01101001011011100111010101111000
00101101011001110110111001110101
00101111011011000110010000101101
01101100011010010110111001110101
01111000001011010111100000111000
00110110001011010011011000110100

...

[illegible]

preprocessing

compiling

assembling

linking

compiling

decompiling

reverse engineering

[illegible]



types

bool

int

long

float

double

char

string

...

bool 1 byte

int 4 bytes

long 8 bytes

float 4 bytes

double 8 bytes

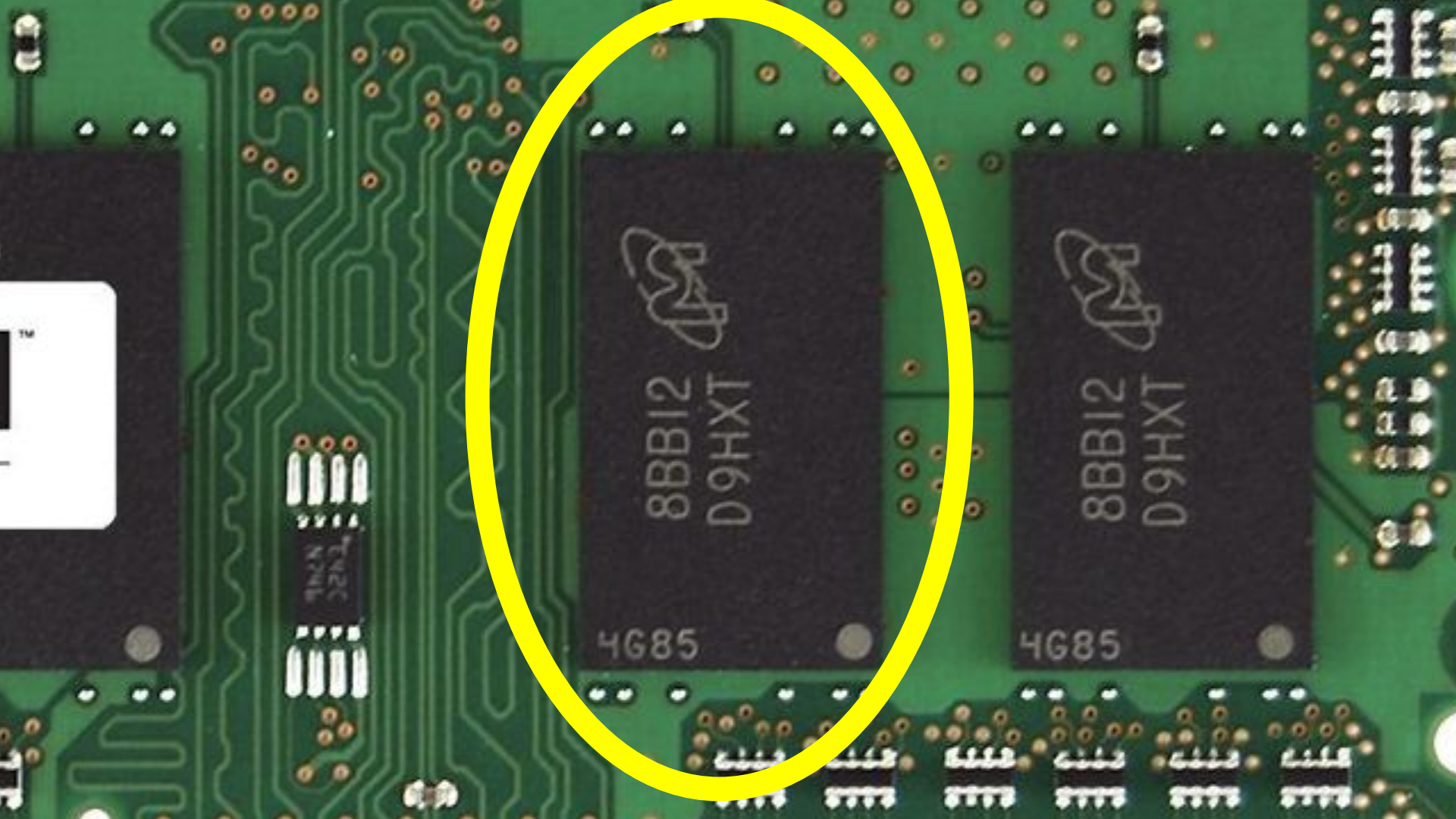
char 1 byte

string ? bytes

...







8BB12
D9HXT

4G85



8BB12
D9HXT

4G85



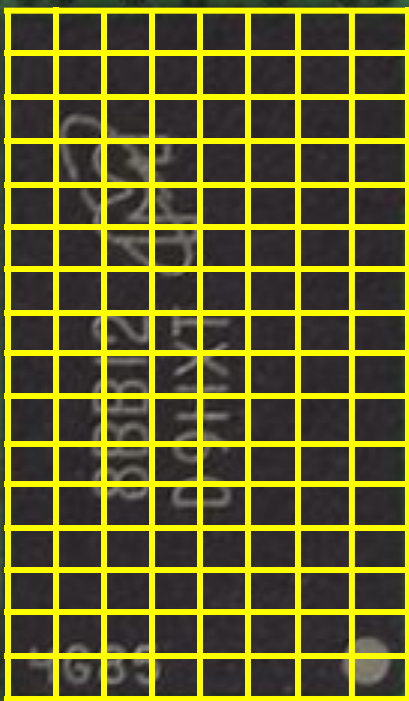
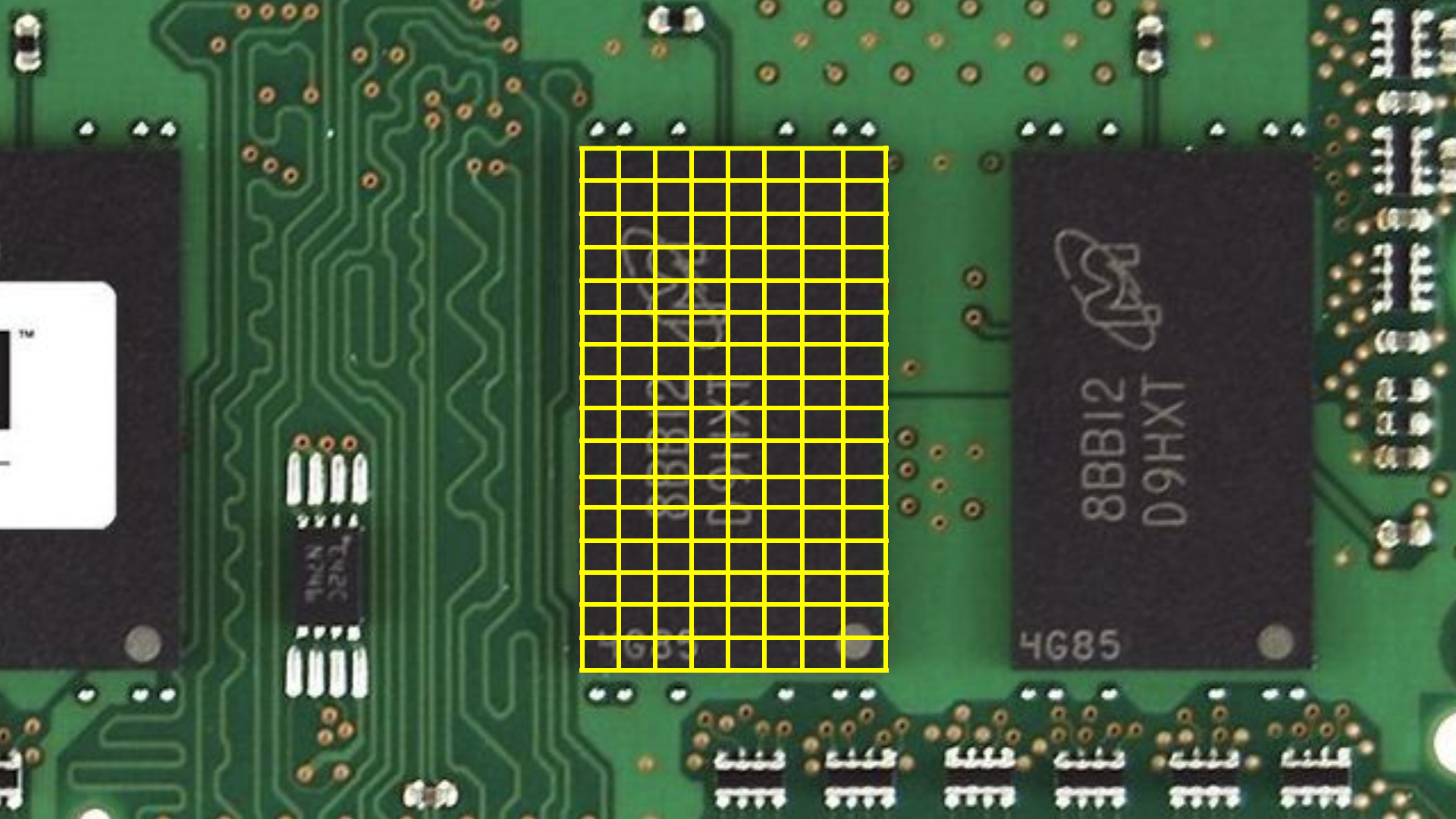
8BB12
D9HXT

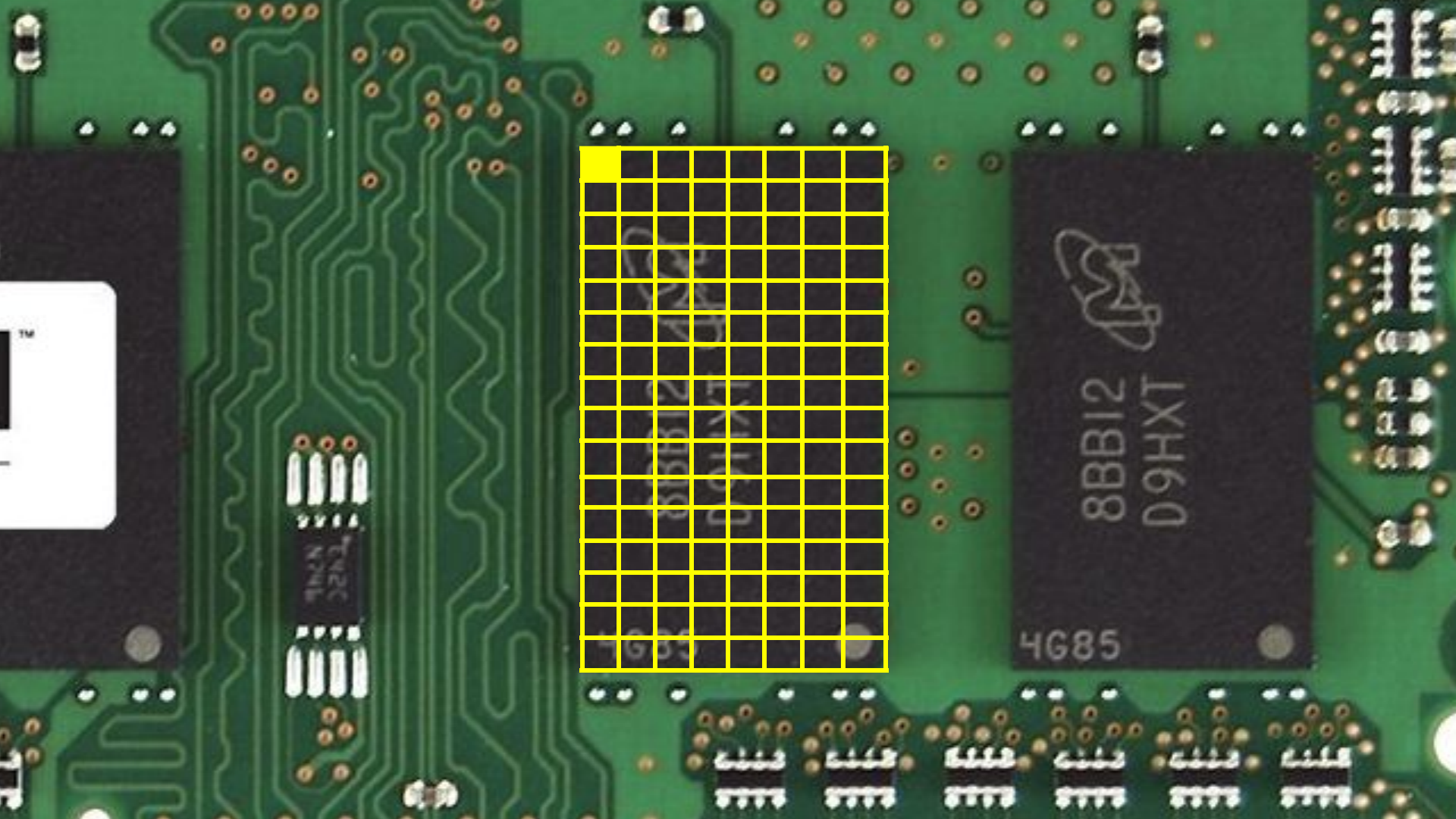
4G85

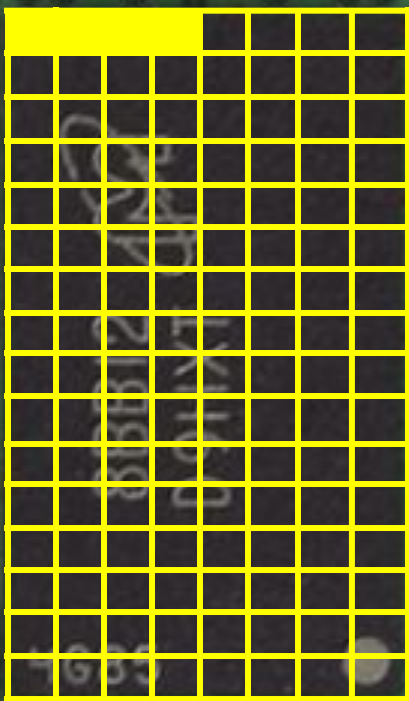
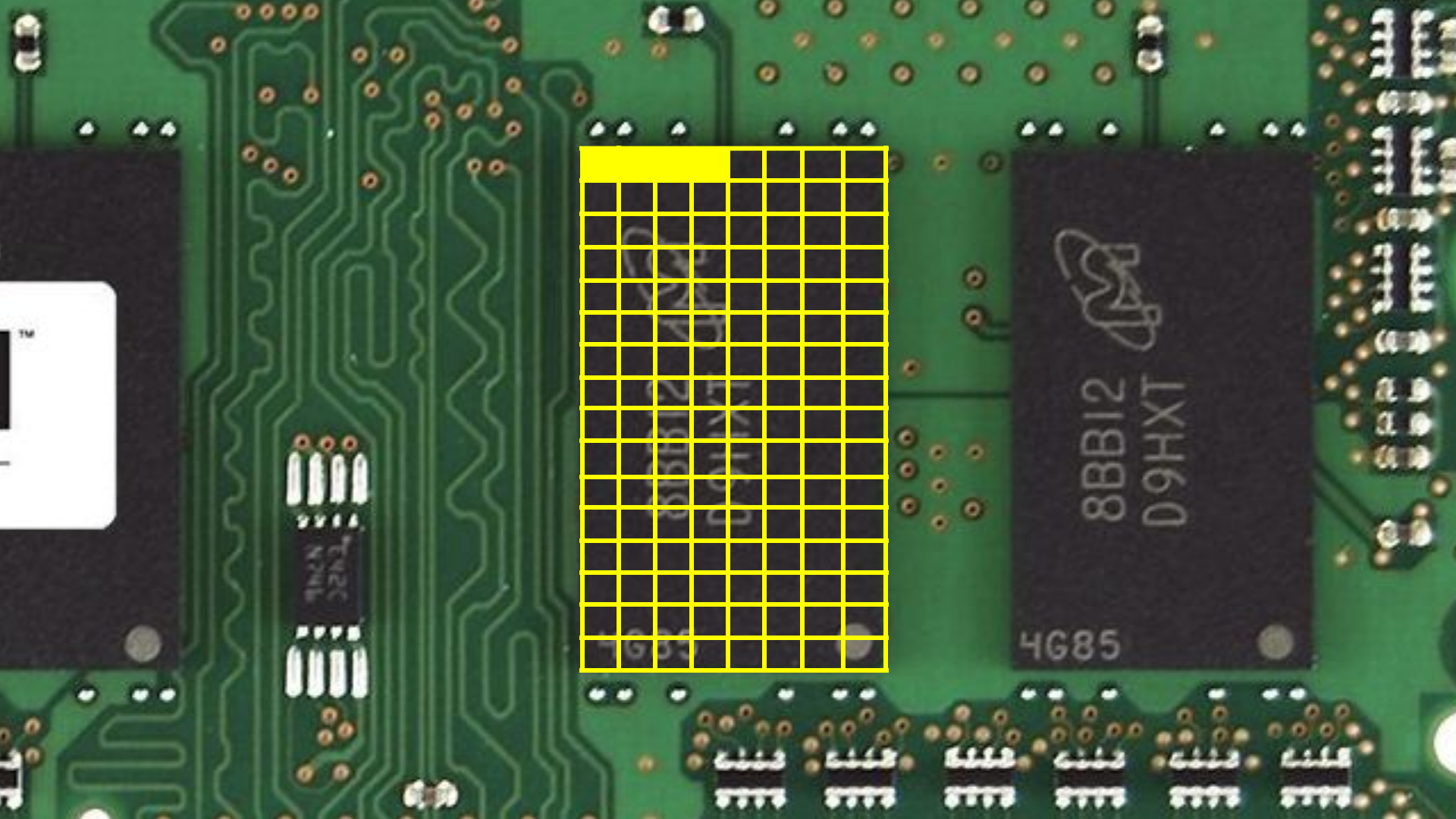


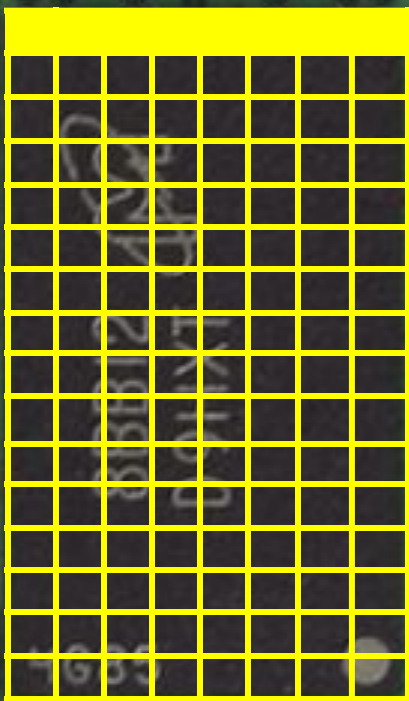
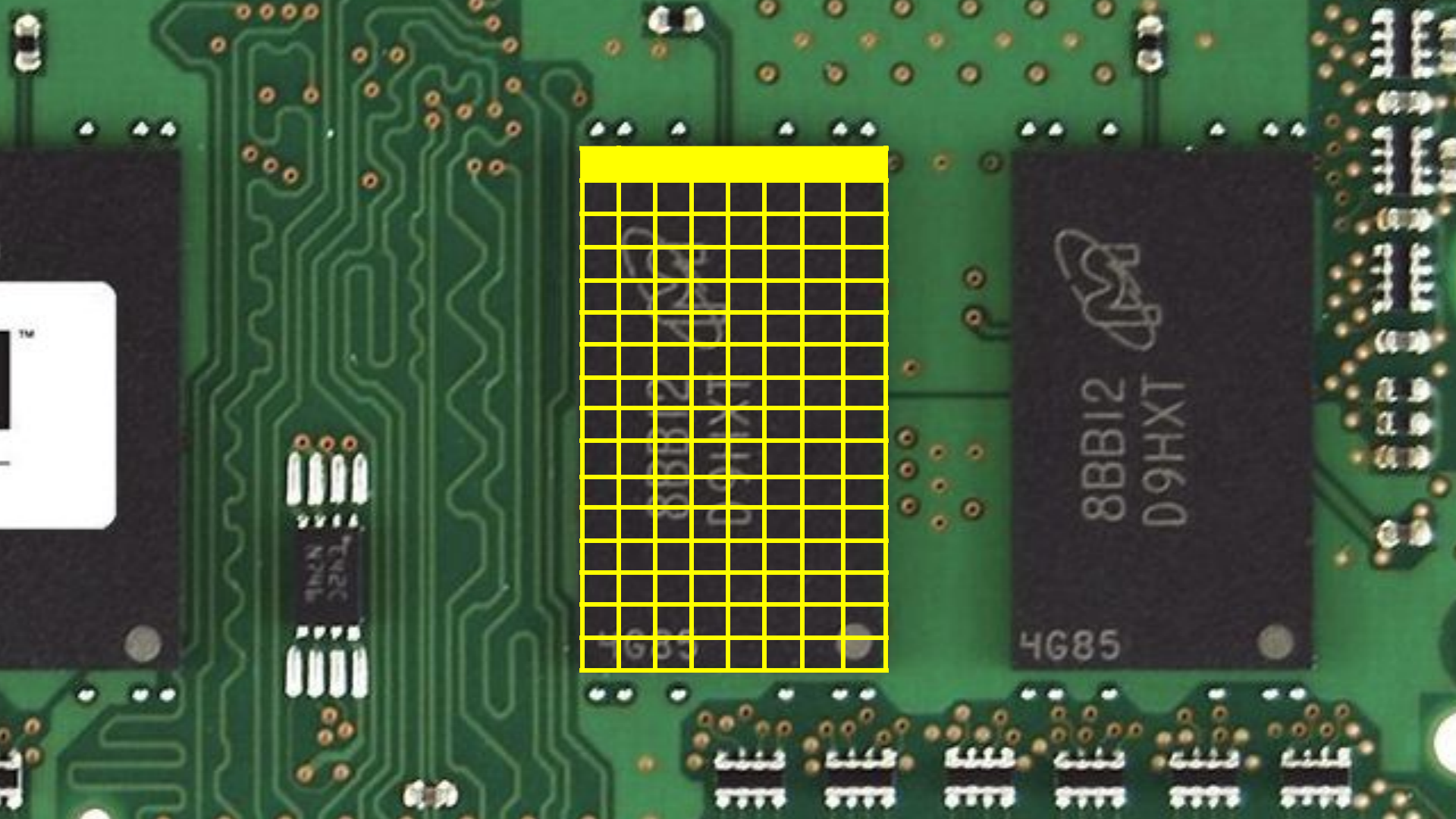
8BB12
D9HXT

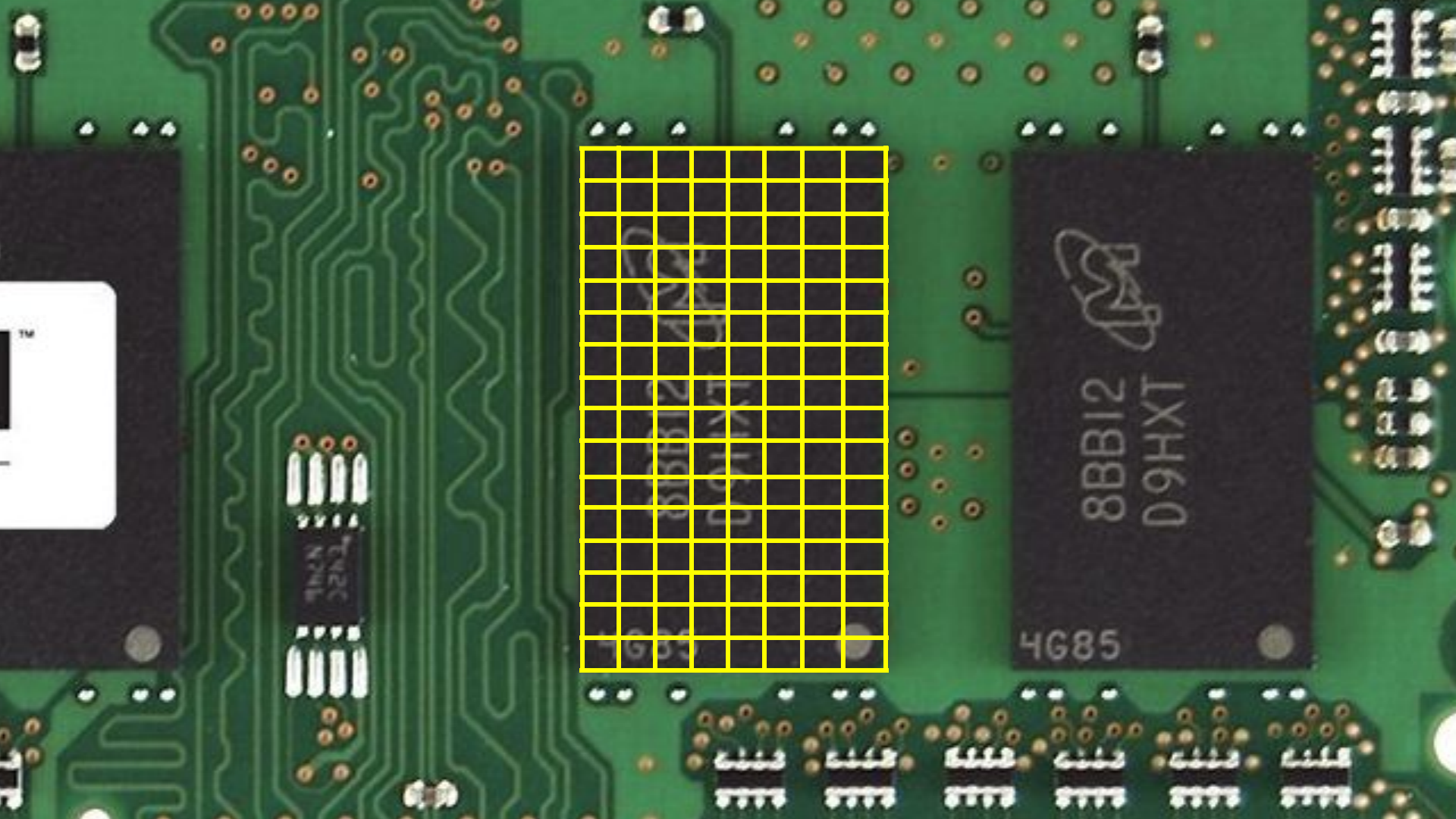
4G85

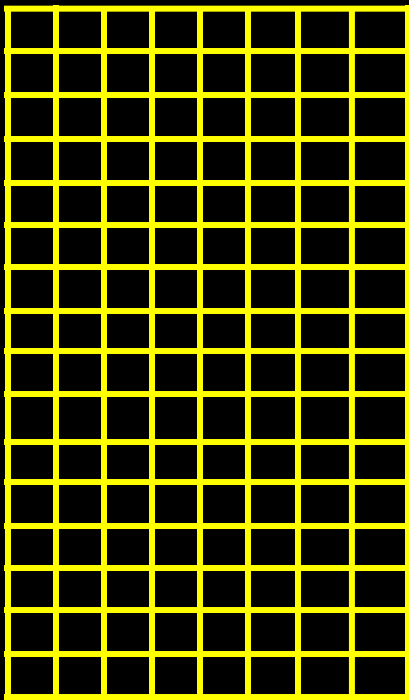


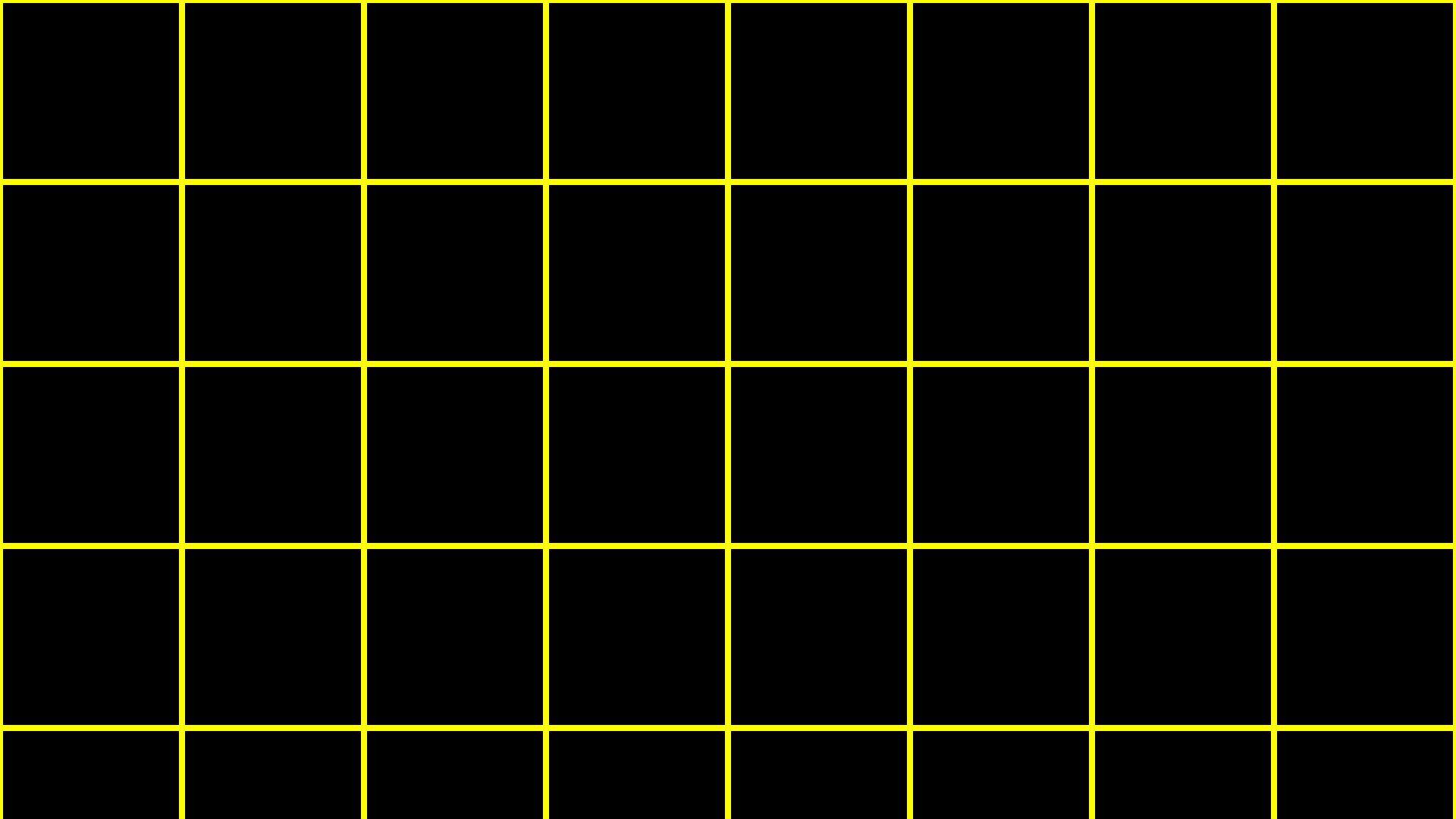








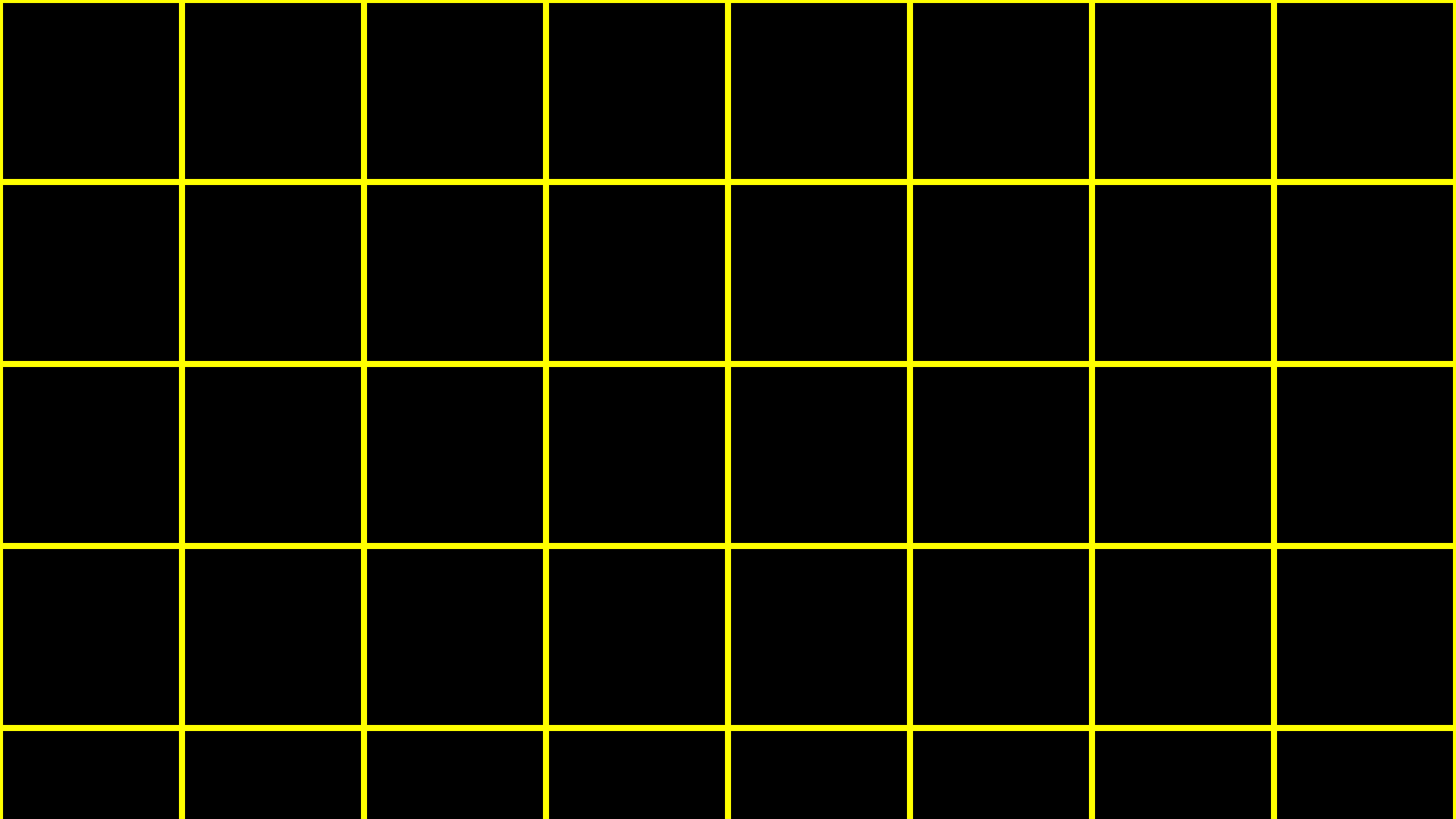




```
int score1 = 72;
```

```
int score2 = 73;
```

```
int score3 = 33;
```



72

score1

72 score1							

72

score1

73

score2

72

score1

73

score2

33

score3

[illegible]

score1

[illegible]

score2

00000000000000000000000000000000100001

score3

```
int score1 = 72;
```

```
int score2 = 73;
```

```
int score3 = 33;
```

arrays

```
int scores[3];
```

```
int scores[3];
```

```
scores[0] = 72;
```

```
scores[1] = 73;
```

```
scores[2] = 33;
```

72

scores[0]

73

scores[1]

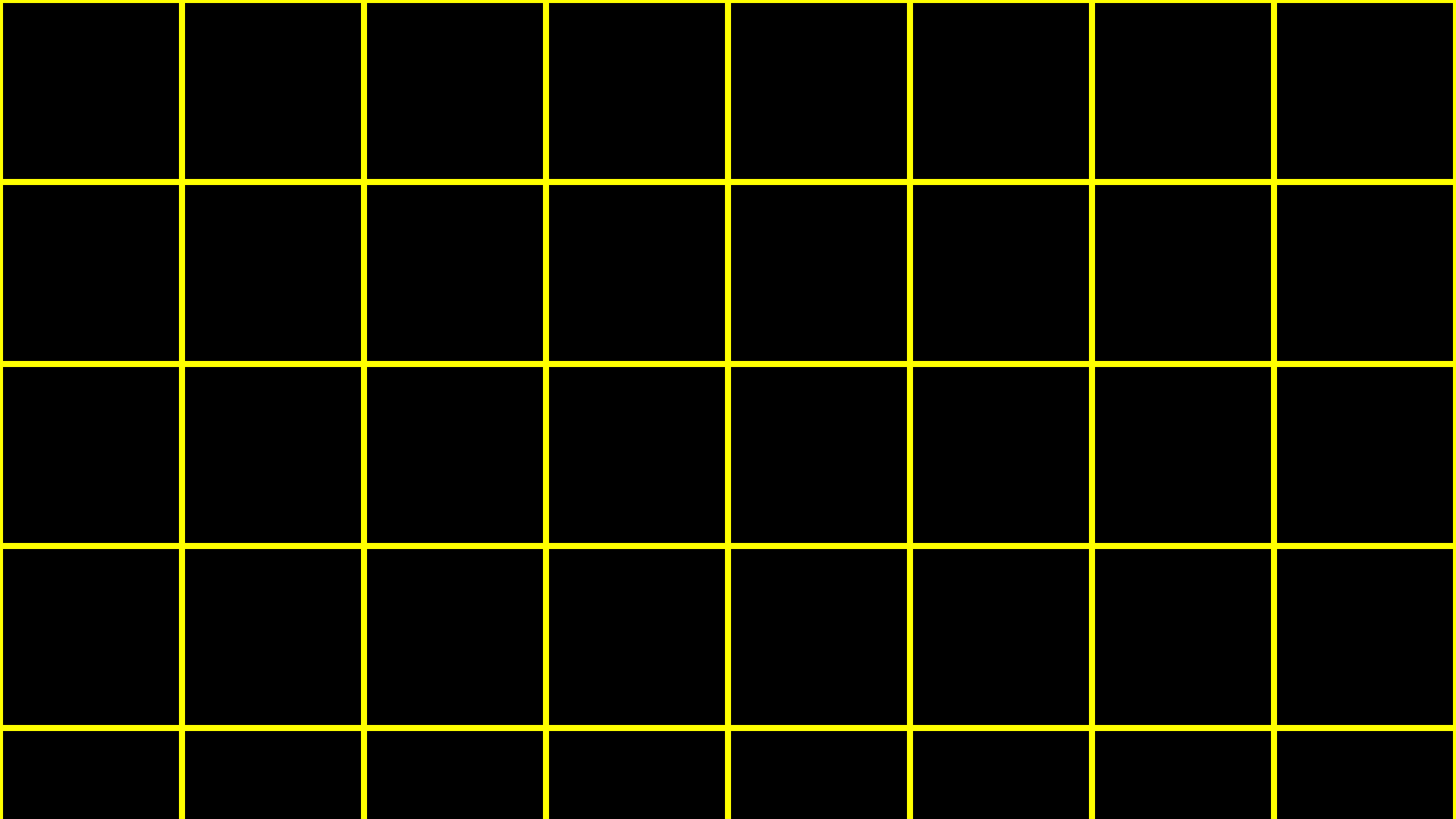
33

scores[2]

```
char c1 = 'H';
```

```
char c2 = 'I';
```

```
char c3 = '!';
```



H

c1

I

c2

!

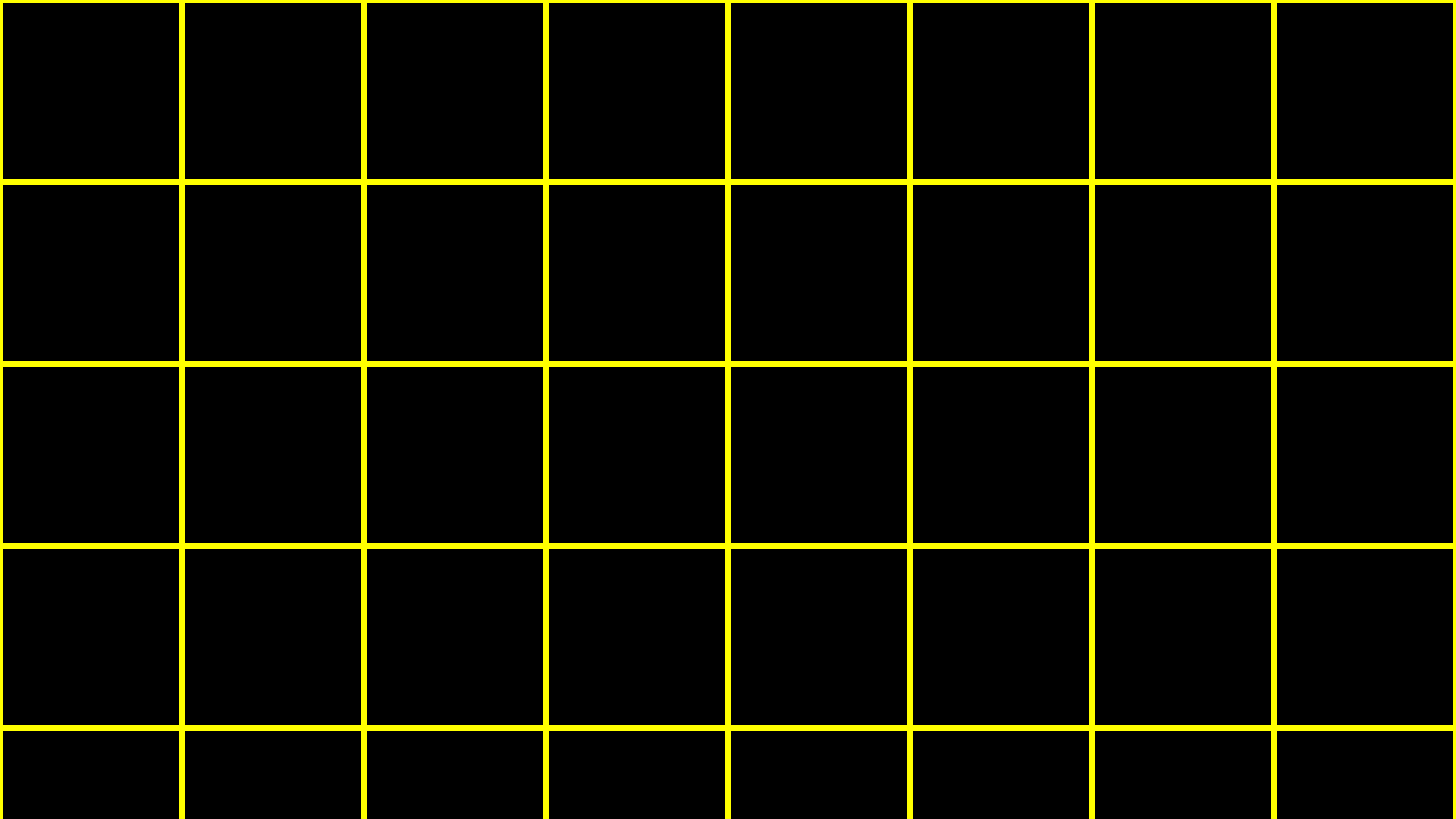
c3

72 c1	73 c2	33 c3					

01001000 c1	01001001 c2	00100001 c3					

string

```
string s = "HI!";
```



H

I

!

s

H s[0]	I s[1]	! s[2]					

H s[0]	I s[1]	! s[2]	00000000 s[3]				

H s[0]	I s[1]	! s[2]	0 s[3]				

H s[0]	I s[1]	! s[2]	\0 s[3]				

72 s[0]	73 s[1]	33 s[2]	0 s[3]				

H

s

I

!

\0

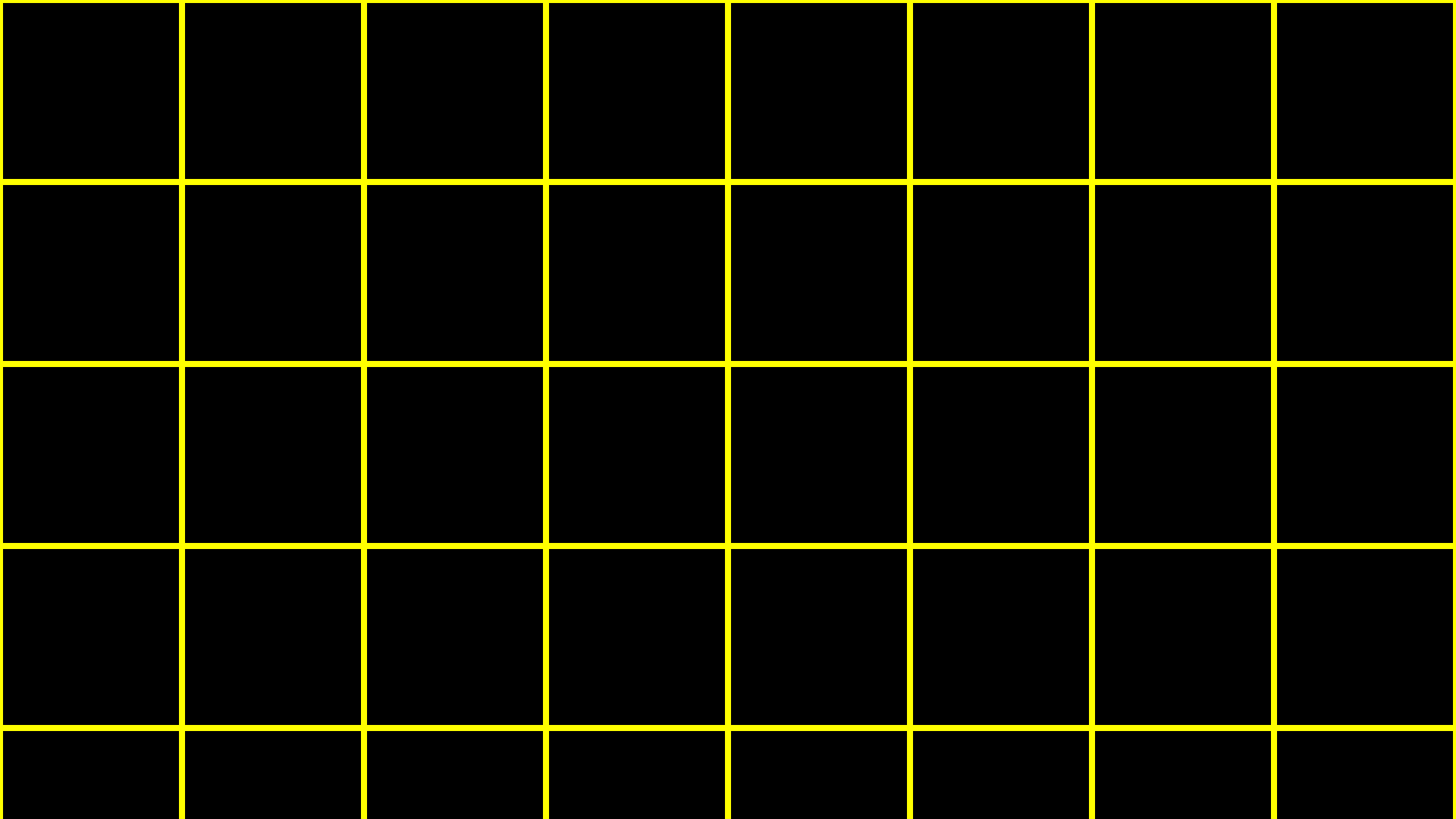
NUL

0	<u>NUL</u>	16	<u>DLE</u>	32	<u>SP</u>	48	0	64	@	80	P	96	`	112	p
1	<u>SOH</u>	17	<u>DC1</u>	33	!	49	1	65	A	81	Q	97	a	113	q
2	<u>STX</u>	18	<u>DC2</u>	34	"	50	2	66	B	82	R	98	b	114	r
3	<u>ETX</u>	19	<u>DC3</u>	35	#	51	3	67	C	83	S	99	c	115	s
4	<u>EOT</u>	20	<u>DC4</u>	36	\$	52	4	68	D	84	T	100	d	116	t
5	<u>ENQ</u>	21	<u>NAK</u>	37	%	53	5	69	E	85	U	101	e	117	u
6	<u>ACK</u>	22	<u>SYN</u>	38	&	54	6	70	F	86	V	102	f	118	v
7	<u>BEL</u>	23	<u>ETB</u>	39	'	55	7	71	G	87	W	103	g	119	w
8	<u>BS</u>	24	<u>CAN</u>	40	(	56	8	72	H	88	X	104	h	120	x
9	<u>HT</u>	25	<u>EM</u>	41	)	57	9	73	I	89	Y	105	i	121	y
10	<u>LF</u>	26	<u>SUB</u>	42	*	58	:	74	J	90	Z	106	j	122	z
11	<u>VT</u>	27	<u>ESC</u>	43	+	59	;	75	K	91	[	107	k	123	{
12	<u>FF</u>	28	<u>FS</u>	44	,	60	<	76	L	92	\	108	l	124	
13	<u>CR</u>	29	<u>GS</u>	45	-	61	=	77	M	93	]	109	m	125	}
14	<u>SO</u>	30	<u>RS</u>	46	.	62	>	78	N	94	^	110	n	126	~
15	<u>SI</u>	31	<u>US</u>	47	/	63	?	79	O	95	_	111	o	127	<u>DEL</u>

0	<u>NUL</u>	16	<u>DLE</u>	32	<u>SP</u>	48	0	64	@	80	P	96	`	112	p
1	<u>SOH</u>	17	<u>DC1</u>	33	!	49	1	65	A	81	Q	97	a	113	q
2	<u>STX</u>	18	<u>DC2</u>	34	"	50	2	66	B	82	R	98	b	114	r
3	<u>ETX</u>	19	<u>DC3</u>	35	#	51	3	67	C	83	S	99	c	115	s
4	<u>EOT</u>	20	<u>DC4</u>	36	\$	52	4	68	D	84	T	100	d	116	t
5	<u>ENQ</u>	21	<u>NAK</u>	37	%	53	5	69	E	85	U	101	e	117	u
6	<u>ACK</u>	22	<u>SYN</u>	38	&	54	6	70	F	86	V	102	f	118	v
7	<u>BEL</u>	23	<u>ETB</u>	39	'	55	7	71	G	87	W	103	g	119	w
8	<u>BS</u>	24	<u>CAN</u>	40	(	56	8	72	H	88	X	104	h	120	x
9	<u>HT</u>	25	<u>EM</u>	41	)	57	9	73	I	89	Y	105	i	121	y
10	<u>LF</u>	26	<u>SUB</u>	42	*	58	:	74	J	90	Z	106	j	122	z
11	<u>VT</u>	27	<u>ESC</u>	43	+	59	;	75	K	91	[	107	k	123	{
12	<u>FF</u>	28	<u>FS</u>	44	,	60	<	76	L	92	\	108	l	124	
13	<u>CR</u>	29	<u>GS</u>	45	-	61	=	77	M	93	]	109	m	125	}
14	<u>SO</u>	30	<u>RS</u>	46	.	62	>	78	N	94	^	110	n	126	~
15	<u>SI</u>	31	<u>US</u>	47	/	63	?	79	O	95	_	111	o	127	<u>DEL</u>

```
string s = "HI!";
```

```
string t = "BYE!";
```



H

s

I

!

\0

H

s

I

!

\0

B

t

Y

E

!

\0

H s[0]	I s[1]	! s[2]	\0 s[3]	B t[0]	Y t[1]	E t[2]	! t[3]
\0 t[4]							

```
string words[2];
```

```
words[0] = "HI!";
```

```
words[1] = "BYE!";
```

H

I

!

\0

words[0]

B

Y

E

!

words[1]

\0

H

words[0][0]

I

words[0][1]

!

words[0][2]

\0

words[0][3]

B

words[1][0]

Y

words[1][1]

E

words[1][2]

!

words[1][3]

\0

words[1][4]

string

string.h

manual.cs50.io/#string.h

strlen

ctype.h

manual.cs50.io/#ctype.h

0	<u>NUL</u>	16	<u>DLE</u>	32	<u>SP</u>	48	0	64	@	80	P	96	`	112	p
1	<u>SOH</u>	17	<u>DC1</u>	33	!	49	1	65	A	81	Q	97	a	113	q
2	<u>STX</u>	18	<u>DC2</u>	34	"	50	2	66	B	82	R	98	b	114	r
3	<u>ETX</u>	19	<u>DC3</u>	35	#	51	3	67	C	83	S	99	c	115	s
4	<u>EOT</u>	20	<u>DC4</u>	36	\$	52	4	68	D	84	T	100	d	116	t
5	<u>ENQ</u>	21	<u>NAK</u>	37	%	53	5	69	E	85	U	101	e	117	u
6	<u>ACK</u>	22	<u>SYN</u>	38	&	54	6	70	F	86	V	102	f	118	v
7	<u>BEL</u>	23	<u>ETB</u>	39	'	55	7	71	G	87	W	103	g	119	w
8	<u>BS</u>	24	<u>CAN</u>	40	(	56	8	72	H	88	X	104	h	120	x
9	<u>HT</u>	25	<u>EM</u>	41	)	57	9	73	I	89	Y	105	i	121	y
10	<u>LF</u>	26	<u>SUB</u>	42	*	58	:	74	J	90	Z	106	j	122	z
11	<u>VT</u>	27	<u>ESC</u>	43	+	59	;	75	K	91	[	107	k	123	{
12	<u>FF</u>	28	<u>FS</u>	44	,	60	<	76	L	92	\	108	l	124	
13	<u>CR</u>	29	<u>GS</u>	45	-	61	=	77	M	93	]	109	m	125	}
14	<u>SO</u>	30	<u>RS</u>	46	.	62	>	78	N	94	^	110	n	126	~
15	<u>SI</u>	31	<u>US</u>	47	/	63	?	79	O	95	_	111	o	127	<u>DEL</u>

0	<u>NUL</u>	16	<u>DLE</u>	32	<u>SP</u>	48	0	64	@	80	P	96	`	112	p
1	<u>SOH</u>	17	<u>DC1</u>	33	!	49	1	65	A	81	Q	97	a	113	q
2	<u>STX</u>	18	<u>DC2</u>	34	"	50	2	66	B	82	R	98	b	114	r
3	<u>ETX</u>	19	<u>DC3</u>	35	#	51	3	67	C	83	S	99	c	115	s
4	<u>EOT</u>	20	<u>DC4</u>	36	\$	52	4	68	D	84	T	100	d	116	t
5	<u>ENQ</u>	21	<u>NAK</u>	37	%	53	5	69	E	85	U	101	e	117	u
6	<u>ACK</u>	22	<u>SYN</u>	38	&	54	6	70	F	86	V	102	f	118	v
7	<u>BEL</u>	23	<u>ETB</u>	39	'	55	7	71	G	87	W	103	g	119	w
8	<u>BS</u>	24	<u>CAN</u>	40	(	56	8	72	H	88	X	104	h	120	x
9	<u>HT</u>	25	<u>EM</u>	41	)	57	9	73	I	89	Y	105	i	121	y
10	<u>LF</u>	26	<u>SUB</u>	42	*	58	:	74	J	90	Z	106	j	122	z
11	<u>VT</u>	27	<u>ESC</u>	43	+	59	;	75	K	91	[	107	k	123	{
12	<u>FF</u>	28	<u>FS</u>	44	,	60	<	76	L	92	\	108	l	124	
13	<u>CR</u>	29	<u>GS</u>	45	-	61	=	77	M	93	]	109	m	125	}
14	<u>SO</u>	30	<u>RS</u>	46	.	62	>	78	N	94	^	110	n	126	~
15	<u>SI</u>	31	<u>US</u>	47	/	63	?	79	O	95	_	111	o	127	<u>DEL</u>

main

```
#include <stdio.h>
```

```
int main(void)
```

```
{
```

```
    ...
```

```
}
```

```
#include <stdio.h>
```

```
int main(void)
```

```
{
```

```
...
```

```
}
```

```
#include <stdio.h>
```

```
int main(int argc, string argv[])  
{  
    ...  
}
```

cowSay

exit status



Zoom



An unknown error occurred

Error code: 1132

[Report Problem](#)

OK

404

This is not the
web page you
are looking for.



```
#include <stdio.h>
```

```
int main(int argc, string argv[])  
{  
    ...  
}
```

```
#include <stdio.h>
```

```
int main(int argc, string argv[])  
{  
    ...  
}
```

```
#include <stdio.h>
```

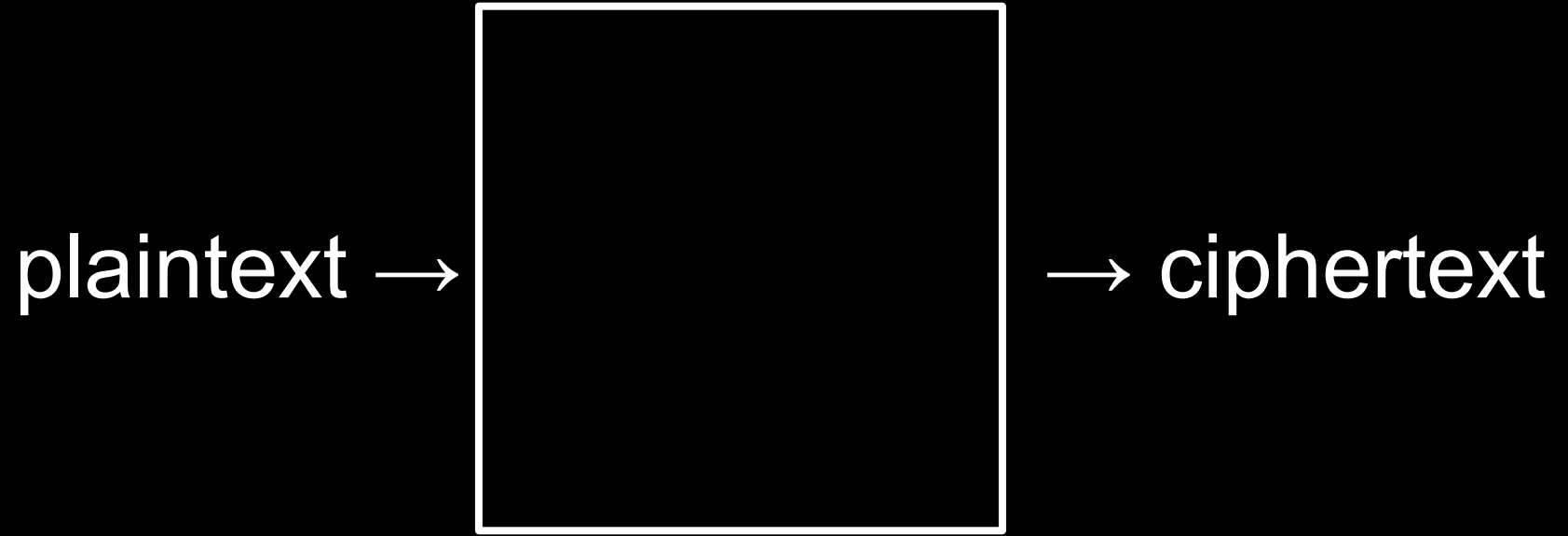
```
int main(void)  
{  
    ...  
}
```

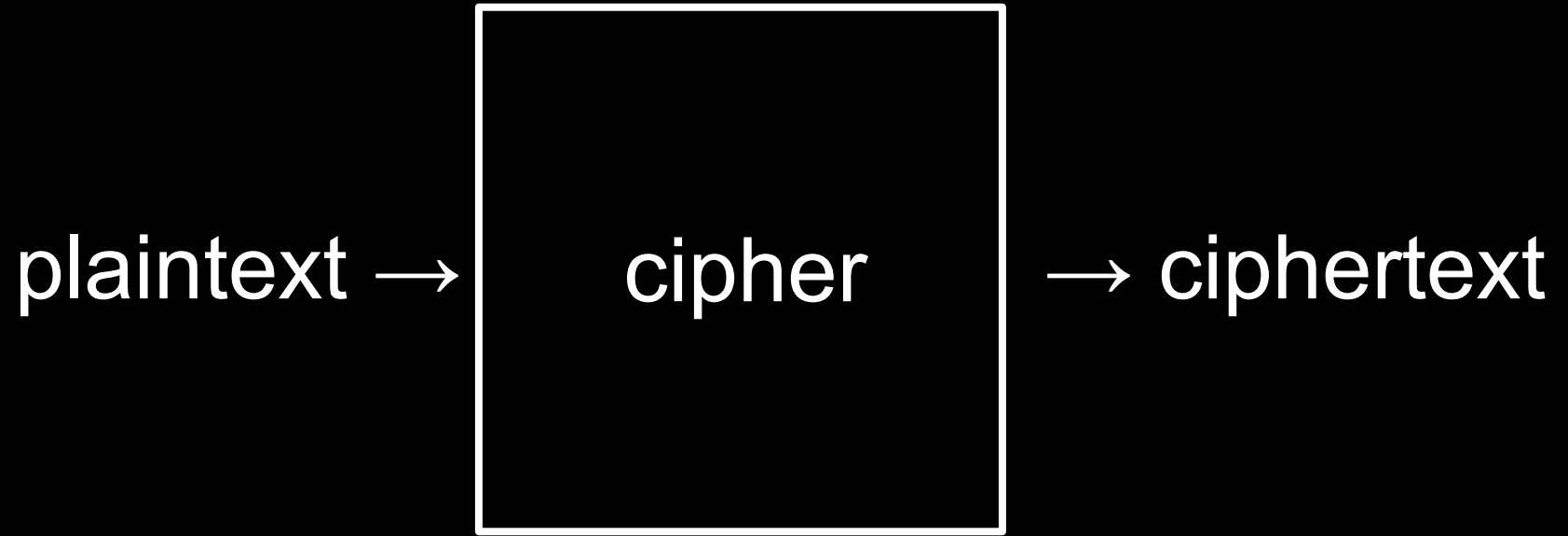
echo \$?

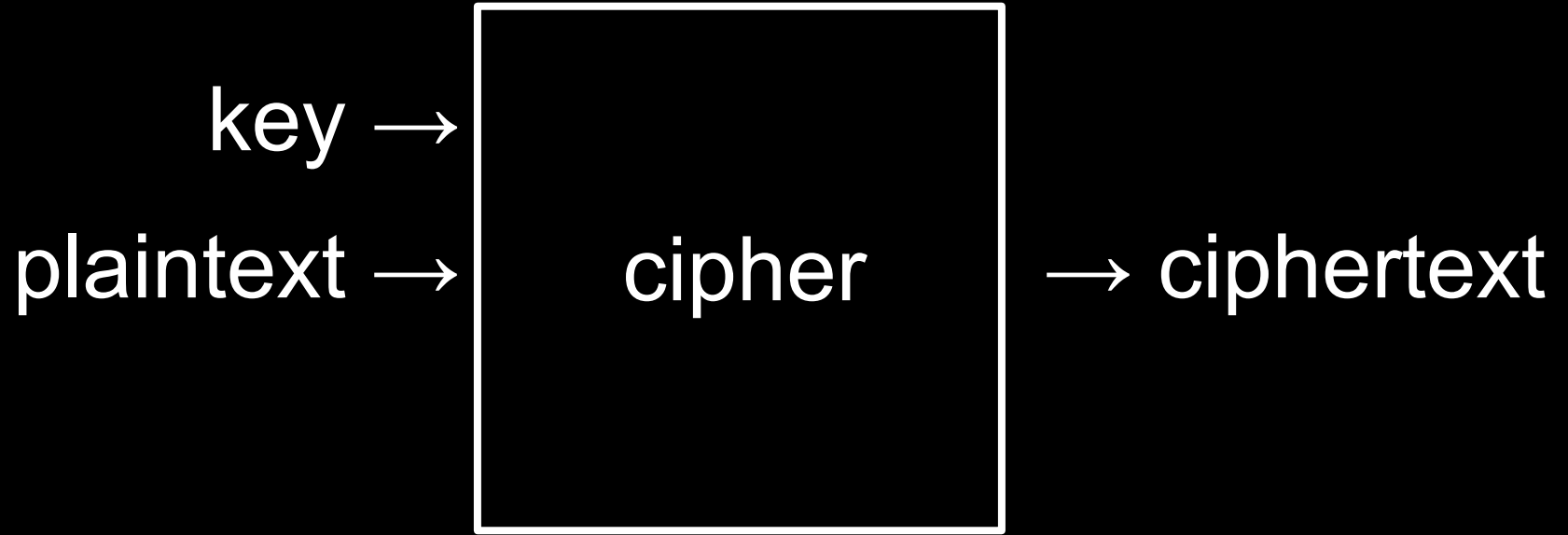
cryptography

encryption



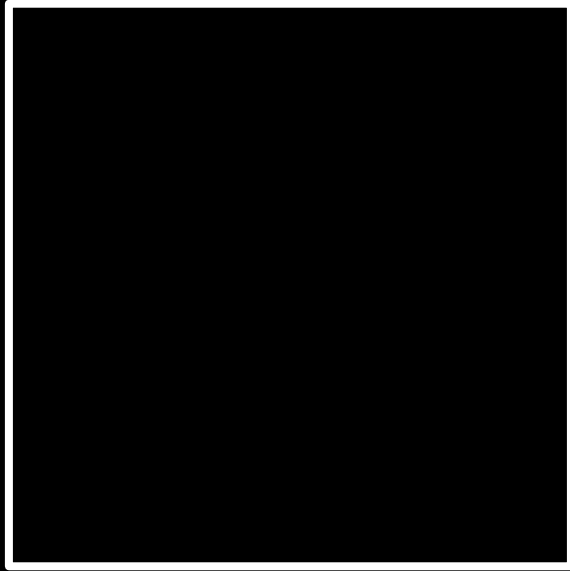


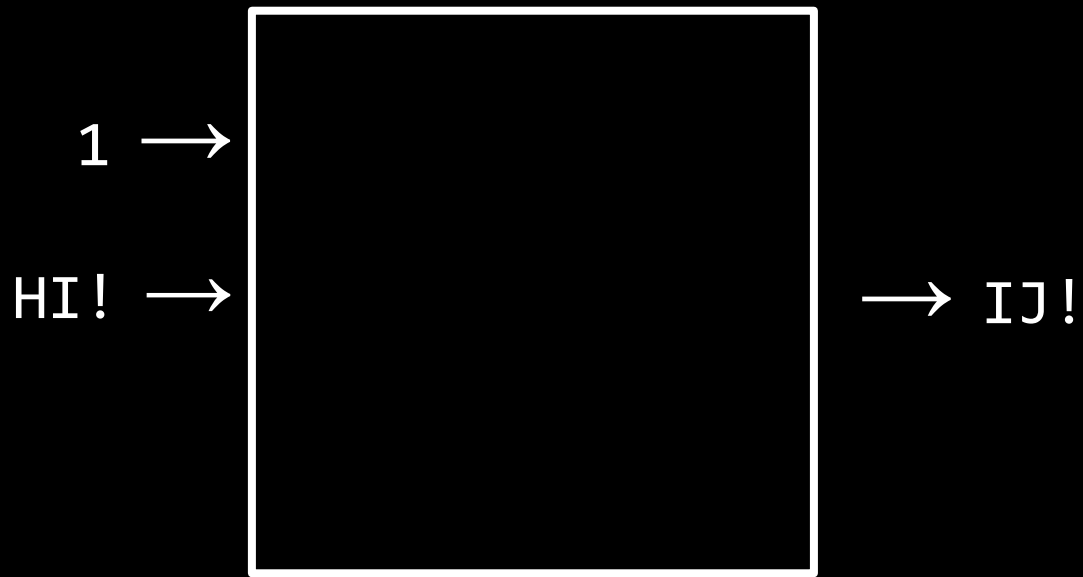


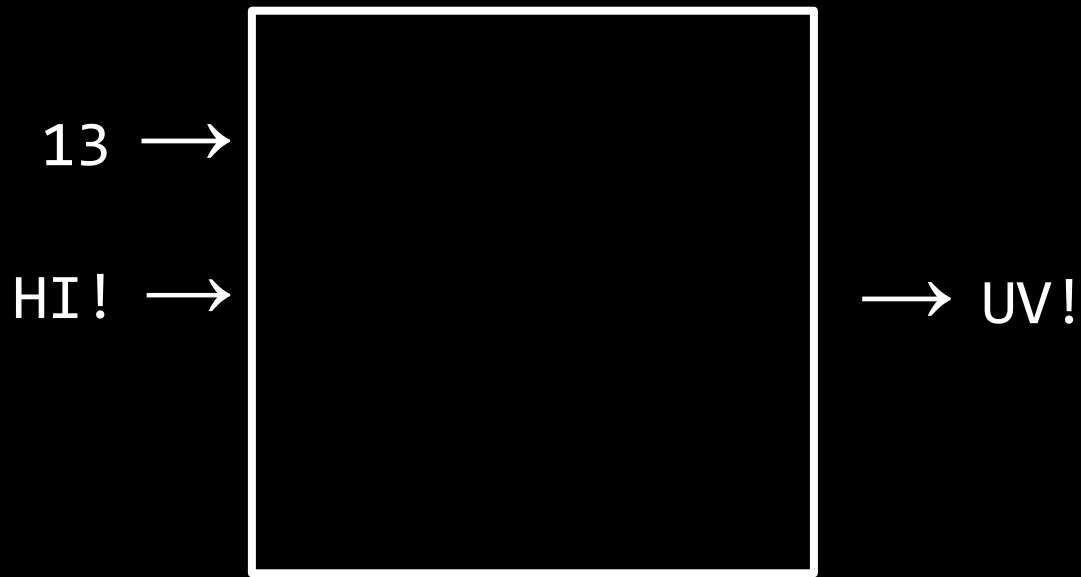


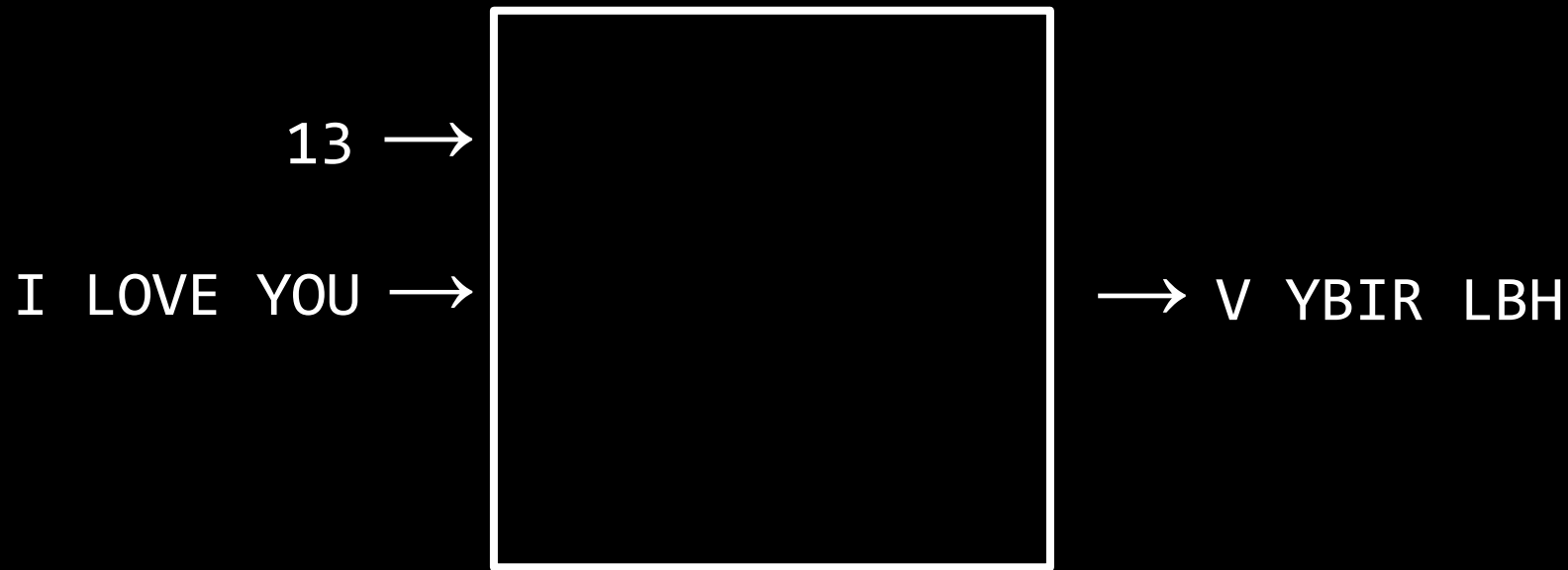
1 →

HI! →



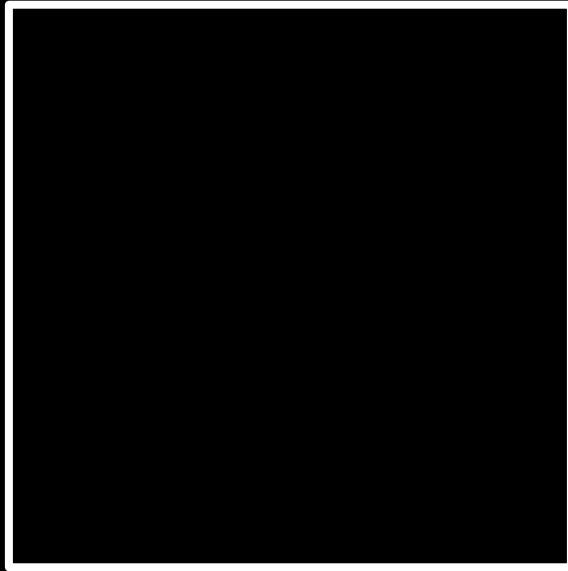






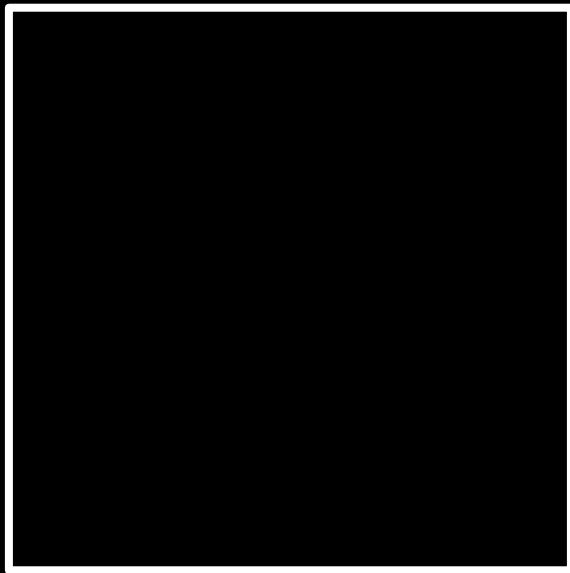
26 →

I LOVE YOU →



26 →

I LOVE YOU →

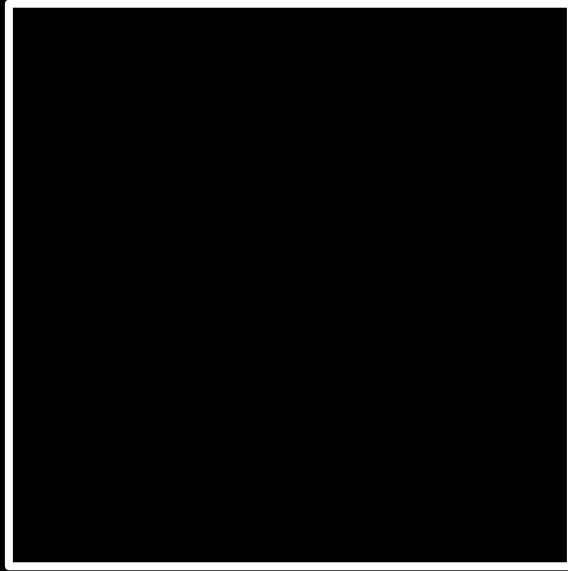


→ I LOVE YOU

decryption

-1 →

UIJT XBT DT50 →



U I J T X B T D T 5 0

T I J T X B T D T 5 0

T H J T X B T D T 5 0

T H I T X B T D T 5 0

T H I S X B T D T 5 0

T H I S W B T D T 5 0

T H I S W A T D T 5 0

T H I S W A S D T 5 0

T H I S W A S C T 5 0

T H I S W A S C S 5 0



This is CS50